collagen microscopy bead modulus rheology tissue spheroid bioprinting polymer elastic rheology coating bioprinting strain granule tissue displacement network elastic fibroblast coating coating coating porosity granule bioprinting polymer strain coating density network tissue rheology diffusion network stiffness network network confocal field coating viscoelastic diffusion elastic adhesion field modulus matrix crosslink strain crosslink migration stress field rheology crosslink scaffold traction spheroid fluorescent scaffold viscoelastic adhesion gradient diffusion gradient hydrogel tissue crosslink elastic cellular density scaffold gradient rheology coating spheroid traction stress scaffold cellular cellular crosslink network granule migration porosity diffusion network scaffold crosslink stiffness stiffness confocal displacement diffusion granule bioprinting crosslink collagen density diffusion polymer strain force spheroid gradient diffusion migration crosslink viscoelastic rheology stiffness viscoelastic stiffness granule porosity porosity matrix confocal coating network adhesion diffusion adhesion hydrogel diffusion bead traction microscopy hydrogel coating tissue granule displacement fluorescent displacement modulus adhesion stiffness field microscopy cellular cellular bead density cellular displacement field confocal fibroblast rheology spheroid modulus coating stress bioprinting matrix viscoelastic migration bead elastic bead crosslink polymer strain coating network coating scaffold substrate traction cellular tissue crosslink strain porosity network density tissue network density coating scaffold fibroblast strain force stress collagen polymer force stress microscopy microscopy stress stress cellular viscoelastic bead collagen granule diffusion traction polymer confocal cellular crosslink traction bioprinting migration stiffness elastic polymer strain migration rheology elastic bioprinting field crosslink rheology coating fibroblast scaffold field coating cellular migration fibroblast collagen matrix strain polymer displacement elastic bioprinting diffusion stiffness porosity rheology porosity fluorescent microscopy traction hydrogel collagen cellular cellular porosity polymer displacement matrix crosslink bead gradient matrix matrix field fluorescent rheology collagen diffusion elastic fibroblast traction viscoelastic microscopy bioprinting substrate collagen matrix modulus bioprinting microscopy diffusion network hydrogel displacement gradient field porosity modulus confocal displacement elastic traction field granule granule hydrogel viscoelastic modulus traction migration fluorescent viscoelastic cellular modulus tissue cellular fluorescent cellular elastic strain bioprinting porosity field diffusion bead spheroid fibroblast elastic polymer fibroblast traction coating granule field fibroblast tissue scaffold fibroblast scaffold microscopy microscopy fibroblast confocal modulus bead polymer porosity spheroid stiffness bead adhesion porosity polymer stress migration fluorescent gradient hydrogel displacement hydrogel tissue hydrogel matrix network bioprinting stress traction fibroblast adhesion fibroblast stress fluorescent matrix elastic porosity hydrogel fluorescent network coating fluorescent scaffold microscopy displacement diffusion microscopy microscopy coating granule field stiffness rheology spheroid substrate elastic crosslink fibroblast microscopy crosslink adhesion adhesion substrate substrate fibroblast stress elastic crosslink field collagen polymer substrate porosity traction fibroblast diffusion polymer adhesion stress strain porosity cellular force fluorescent bead microscopy tissue strain diffusion bead porosity tissue porosity confocal granule scaffold matrix cellular bead rheology coating viscoelastic coating force stiffness collagen collagen collagen bead displacement scaffold scaffold adhesion hydrogel network rheology granule adhesion density fibroblast crosslink tissue network fluorescent fibroblast rheology spheroid network viscoelastic matrix diffusion displacement network force microscopy crosslink gradient cellular crosslink polymer stress stress stress diffusion gradient cellular confocal hydrogel modulus crosslink bioprinting adhesion substrate bead strain polymer force rheology scaffold stiffness bioprinting crosslink cellular porosity traction density hydrogel bead elastic displacement hydrogel collagen hydrogel tissue fluorescent bioprinting strain scaffold cellular fibroblast tissue collagen rheology